



Dysphagia diagnosis system with integrated speech analysis from throat vibration[☆]

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ABSTRACT

Dysphagia is a widespread swallowing disorder. One of the most widely used clinical assessments is the bedside swallowing disorder screening, including the water swallow test. This method can only diagnose the disease according to the choking situation of the liquid, and the aspiration may appear during the drinking water test. In this paper, an automatic diagnostic system is designed according to the patient's performance in pronunciation. To diagnose dysphagia, we treat it as a binary classification task. We use a throat vibrator to collect the vibration signals of the subjects during pronunciation for clear speech signals. Feature engineering techniques, including feature extraction, data balancing and dimensionality reduction are implemented to extract the features for classification. Three popular classifiers are integrated to make the final predictions. Experimental results show that the integrated classifier performs better with accuracy = 72.09%, sensitivity = 63.64% and specificity = 80.95% respectively. Different from the traditional clinical ways, a new machine learning based swallowing disorders detection system is proposed with vibration signals of throat acquired for high-quality speech analysis. It may help to realize a portal and low-cost telemedicine device for dysphagia diagnosis at home.

1. Introduction

Dysphagia (also called swallowing disorders) is the difficulty of swallowing due to impaired function of the nerves or organs associated with swallowing (Cheng, Sasegbon, & Hamdy, 2022; Guénard-Lampron, Masson, & Blumenthal, 2021). It causes a series of complications (Selvanderan, Wong, Holloway, & Kuo, 2021), such as aspiration pneumonia and psychological disorders, which seriously affect the quality of patients' life. It is very prevalent in the older population (Christmas & Rogus-Pulia, 2019; Thiyyagalingam, Kulinski, Thorsteinsdottir, Shindelar, & Takahashi, 2021), and it is considered as a geriatric syndrome (Baijens et al., 2016). The prevalence of dysphagia among older population in the community is 27% (Serra-Prat et al., 2011). It increases to 47.5% in hospitalized older patients in an acute geriatric unit (Carrión et al., 2015), and rises to 91% in

older patients hospitalized with community-acquired pneumonia (Almirall et al., 2013). Besides, dysphagia is a usual symptom in many neurological disorders, such as acute stroke (Smithard et al., 1997), Alzheimer's disease (AD) (Horner, Alberts, Dawson, & Cook, 1994; Langmore, Olney, Lomen-Hoerth, & Miller, 2007) and Parkinson's disease (PD) (Lahmiri, Dawson, & Shmuel, 2018; Miller et al., 2009; Volonte, Porta, & Comi, 2002; Walker, Dunn, & Gray, 2011). The incidence of acute stroke complicated by dysphagia can reach up to 78% (Serra-Prat et al., 2011), and the prevalence of swallowing disorders in PD patients is up to 82% (Kalf, De Swart, Bloem, & Munneke, 2012). Despite this high prevalence among the older population and neurological disorders patients, there is no convenient and efficient diagnostic system for home or community to find highly risky subjects for further diagnosis and treatment.

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The gold standard of dysphagia diagnosis is Videofluoroscopic swallowing study (VFSS) (Hutchison, Wishart, Brown, Ward, & Porceddu, 2021). It can clearly assess and diagnose the condition in swallowing therapy. Most of the previous studies (Bülow, Olsson, & Ekberg, 2005; Chang et al., 2003; Chen et al., 2004; Han, Paik, & Park, 2001) have detected dysphagia based on this method. However, VFSS involves radiation and needs the support of hospital (Taubert, Burns, Ward, & Bassett, 2022). This method is invasive and difficult to popularize and implement, and it requires professionals to judge according to imaging findings. Therefore, drinking water test (Ikuko Okuni, 2022) is usually used to determine swallowing disorders in clinical. The shortcoming of drinking water test is that it can only judge the choking situation of the liquid, and some elderly people may appear aspiration. A safe dysphagia detection system at home is really in need.

Swallowing and generating sound both involve the muscles of the oropharynx (Suzuki et al., 2022). The swallowing disorders might cause abnormal movements of articulators in the speech production process. Studying the speech characteristics of patients with dysphagia may help to identify swallowing disorders. Currently, some works begin to detect dysphagia based on speech analysis. Farneti (2017) studied the connection between voice and dysphagia. Santamato et al. (2009) used a microphone and a notebook computer system to analyze the sounds of swallowing acoustically. They claimed this technique could be incorporated into bedside evaluation. López-De-Ipiña, Satue-Villar, Faundez-Zanuy, Arreola, and Calvo (2016) used the significance test and found that features pitch, HNR, and permutation entropy of speech had significant differences between healthy group and dysphagia group. They designed a dysphagia detection system in further work (López-De-Ipiña et al., 2018). Kang, Kim, Jee, Jo, and Koo (2018) analyzed the acoustic voice parameters of a sustained vowel /a/ for prior to and following swallowing to assess the possibility of aspiration risk. Roldan-Vasco, Orozco-Duque, Suarez-Escudero, and Orozco-Arroyave (2021) achieved automatic detection of functional oropharyngeal dysphagia based on some typical speech features. These previous works provide some references on extracting effective speech features and detecting the dysphagia. However, most of these methods need subjects to complete the swallowing action, which exists risks for some older people and is not convenient and fast enough. In addition, these studies do not focus on investigating early signs of dysphagia. The experiments on screening common elderly people with dysphagia are not conducted. The throat signals collected by throat vibrator which might contain crucial speech details of dysphagia are also not studied in the existing research. Inspired by this, we try to identify people with swallowing disorders by variation in their pronunciation. Our previous work investigated this problem with adaptive boosting (Wang et al., 2021) and we have achieved automatic detection of the Parkinson disease based on speech signal processing technique (Ali, Zhu, Zhang, & Liu, 2019). Exploring automatic detection system for such disease can help relieve the burden of clinicians, and collecting speech data is easily accepted by the patients.

In recent years, machine learning (ML) has widely applied to biomedical engineering (Park, Took, & Seong, 2018). The most common application is ML based medical imaging analysis, e.g., Iqbal, Khan, Saba, and Rehman (2018) concluded recent researches on brain tumor multi-class classification using magnetic resonance imaging. Lee et al. (2006) designed a radial basis classifier to automatically classify aspiration. Dankovičová, Sovák, Drotár, and Vokorokos (2018) utilized ML methods to recognize pathological speech, particularly dysphonia. Yanase and Triantaphyllou (2019) did a comprehensive survey on computer-aided diagnosis (CAD) in medicine of 251 publications. That means based computerized technique medicine become maturer. Besides, telemedicine and remote health monitoring make our life more convenient. Teixeira, Félix, Presedo, and Zamarrón (2013) designed a distributed platform for the home supervision of multiple diseases, which can help patients to autonomously manage their conditions.

Table 1

Definition of the notations in equations.

Notations	Definitions
$\mathbf{X}$	Training samples matrix
$\mathbf{D}$	Linear transform
$\tilde{\mathbf{X}}$	Matrix after dimensionality reduction
K	Reduced dimension
$\mathbf{x}_n$	n th training sample
y_n	n th class label
$\mathbf{w}$	Projection vector
b	Optimization bias of SVM
ξ	Vector of slack variables
$\phi(\cdot)$	Kernel method
κ	Kernel function
y_{pred}	Predicted label
α_k	Weight factor
h_k	Prediction result of k th base classifier
e_k	Classification error of k th base classifier
H	Final classifier of Adaboost
Z_k	Normalization factor
Acc	Accuracy of classification task
Sen	Sensitivity of classification task
Spe	Specificity of classification task

We design an automatic dysphagia detection system based on ML. It needs no swallowing and less testing time, which is more convenient and efficient compared with the other methods. It may help to complete dysphagia diagnosis at home and be added into some already well-established telemedicine systems to remotely analyze the pronunciation data. This system consists of a new speech signal acquisition equipment which obtains the vibration signals of subjects during speech, and an integrated classifier which synthesizes the output of base classifiers.

The main contributions of this work are summarized as follows:

1. A new automatic screening system of dysphagia is proposed. This system achieves an early screening of dysphagia for common elderly people at home, and it could provide some help for the clinicians to diagnose the dysphagia in hospitals.
2. A new group of speech features is tested. In this work, we use some efficient feature engineering methods to extract representative speech features to relate the dysphagia symptom. The results indicate that the group of these features is effective in the dysphagia detection task.
3. A new data acquisition system (throat vibrator) is adopted to collect clear throat signals. This acquisition system is different from the microphone used in the related work. The throat vibrator is refitted from a bone conduction headset and the signals in records carry less noise than the microphone.
4. A new integrated classifier is investigated based on three popular classification models. This integrated classifier can self-adaptively synthesize the performances of all base classifiers to make a final decision. Experimental results show that the proposed classifier performs well in the classification tasks.

The rest of the paper is organized as follows. Section 2 introduces the details of the proposed approach, including the feature engineering methods and the new integrated classifier. Section 3 is about experimental settings, results, and discussions. Finally, we give a conclusion for this paper in Section 4.

2. Proposed method

2.1. Data acquisition

In the proposed method, a new data acquisition system, several effective feature engineering methods, and a new integration scheme of classifiers are combined. The overview of this proposed method is given in Fig. 1. The notations in the equations of the proposed approach are

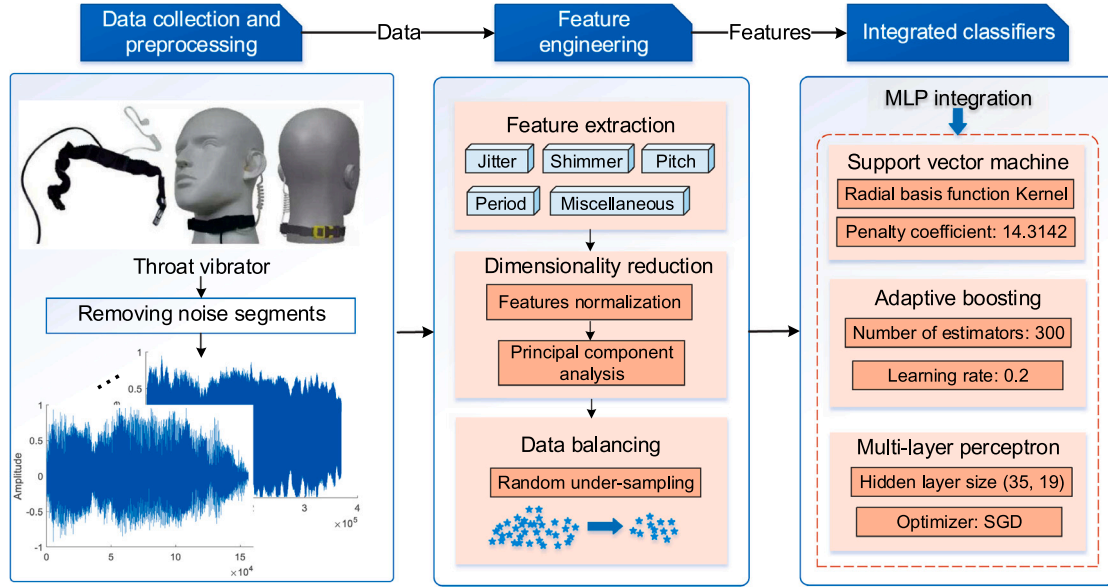


Fig. 1. The diagram of the proposed method.



Fig. 2. Throat vibrator (Wang et al., 2021).

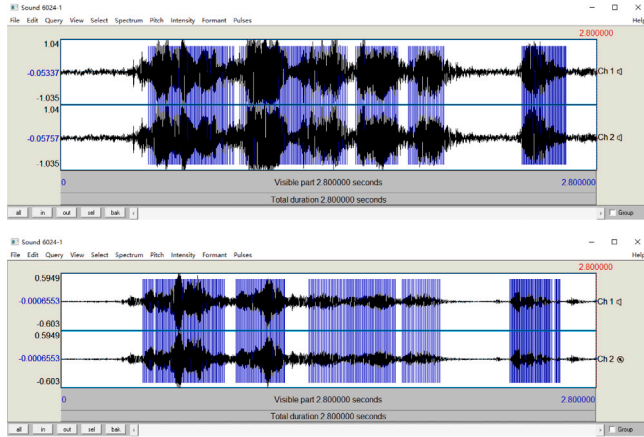


Fig. 3. Comparison between the signals collected by refitted vibration sensors and wearable microphones. The above one is obtained by wearable microphones and the bottom is obtained by refitted vibration sensors (Wang et al., 2021).

introduced in Table 1. The pseudocode code of this method is listed in Table 2.

In Table 2, $\tilde{X}_{train}$ and $\tilde{X}_{test}$ are the matrices of feature set calculated from training and testing samples, respectively. In Train^* function, the first parameter decides on the type of the trained classifier. The second parameter in $[*]$ contains the features adopted for training. $P_{svm-train}$ and $P_{ada-train}$ represent the predicted labels of the training

Table 2

The pseudocode of the proposed dysphagia detection system.

Algorithm: Dysphagia detection system

Input: Throat signals $S = [s_1; s_2; \dots; s_N]$

Initialization: model training function Train^* , feature extraction function F^* , linear transform D ,

$D \in \mathbb{R}^{K \times 26}$, $1 \leq K \leq 26$

1. *Feature extraction*

$X = F(S)$, $X = [x_1; x_2; \dots; x_N] \in \mathbb{R}^{26 \times N}$

2. *Dimensionality reduction*

$\tilde{X} = DX \in \mathbb{R}^{K \times N}$

$\tilde{X} = \tilde{X}_{train} \cup \tilde{X}_{test}$

3. *Dataset balancing*

4. *Integrated classifiers*

$SVM = \text{Train}(svm, [\tilde{X}_{train}])$

$Adaboost = \text{Train}(adaboost, [\tilde{X}_{train}])$

$P_{svm-train} = SVM(\tilde{X}_{train})$

$P_{ada-train} = Adaboost(\tilde{X}_{train})$

$MLP_{int} = \text{Train}(mlp, [\tilde{X}_{train}, P_{svm-train}, P_{ada-train}])$

$P_{svm-test} = SVM(\tilde{X}_{test})$

$P_{ada-test} = Adaboost(\tilde{X}_{test})$

$P_{int-test} = MLP_{int}([\tilde{X}_{test}, P_{svm-test}, P_{ada-test}])$

samples obtained from the trained SVM and Adaboost classifiers, respectively; while $P_{svm-test}$ and $P_{ada-test}$ are the prediction results of the testing samples by these two classifiers. MLP_{int} represents the integrated MLP classifier, and $P_{int-test}$ denotes the prediction results of this classifier.

The acquisition of speech signals is usually carried out through a microphone in the quiet environment. However, data obtained in this way inevitably gets contaminated with noise. In order to overcome this shortcoming and obtain more clean and clear audio data, we adopt a new data acquisition system to replace the traditional microphone. This new system collects speech data by bone conduction, which can not only effectively prevent environmental noise interference, but also fundamentally solve the problem that walkie talkie cannot communicate in a high noise environment. This system uses sensors worn around the throat to capture the vibrations in the vocal cords, is called a laryngeal bone conduction headset. Fig. 2 shows the laryngeal bone conduction headset we used, PTE-796.

In order to better illustrate the advantages of this acquisition system, we compared the throat vibration signals collected by this system with the voice signals collected by the wearable microphone. The two signals

are collected from a same subject simultaneously. Fig. 3 compares the waveforms of these two signals. We find that there are a lot of noises in the speech signal collected by the wearable microphone, which is unfavorable to the subsequent feature engineering. By contrast, the signal picked up by the laryngeal vibrator is more pure.

Based on this new acquisition system, vowels /a/ of each participant are collected. Vowels are sounds made unhindered by air flow through the mouth during pronunciation. In the data collection process, the subjects are asked to pronounce the vowels as long as possible. These audio files are recorded with direct digital 16-bit sampling at a sampling rate of 48.0 kHz.

2.2. Feature engineering

(1) Feature Extraction

In this work, the speech features are extracted based on the collected vowels. Since the collected vowels vary in length and there will be some noise mixed in at the beginning and end of recording, we crop the original audio data and cut out a part of the sound that is relatively stable and smooth. All the audio data is clipped to a length of 2 s.

Next, we take the step of feature extraction for each sample, so as to obtain representative digital features from the original audio signal and then form the feature matrix. 26 standard linear features are extracted from speech signals using the open source software Praat (Boersma, 2006). These features reflect the local and global characteristics of throat vibration signal, including jitter, frequency, period, pitch and so on. Detailed descriptions and statistics of these features can be found in Table 3.

We have analyzed the mean value of each feature in the table and found two issues. First, the values of each feature vary greatly and are not on the same scale, varying from 10^{-3} to 10^3 . Second, there are some features that have different degrees of influence on the detection of dysphagia. Taking the feature Pitch (mean) and Jitter (local) for example, the feature Pitch (mean) represents the average fundamental frequencies of the signal, whose values are usually distributed around 200 Hz. While the feature Jitter (local) represents the frequency perturbation percentage, which are usually around 0.3%. Obviously, the values of Pitch (mean) are much larger than those of Jitter (local). In this case, the classifier tends to learn from features with large values. However, the truth is that healthy people and patients with dysphagia have greater differences in Jitter (local) feature. It indicates the Jitter (local) feature is more important in the classification task.

These extracted 26 audio features represent different physical meanings and have different measurement scales. It is necessary to carry out a normalization processing step. In order to solve the numerical differences of different features, the mean variance normalization method is adopted in this step. For each feature of the test data, we normalize it by removing the mean value of each feature and dividing non-constant features by their standard deviation. It should be noted that the mean and variance here are calculated from the data of the training subset. All the features thus processed will conform to the standard normal distribution.

(2) Dimensionality reduction

Principal component analysis (PCA) technique is used to reduce the dimension of the extracted features. PCA finds out the projection direction of its principal components, and the dimensionality reduction projection is performed on the original data to obtain K main features. It has two advantages: firstly, it eliminates redundancy and improves accuracy; secondly, it reduces feature dimension and avoids overfitting problems.

For the training samples $\mathbf{X} = [\mathbf{x}_1; \mathbf{x}_2; \dots; \mathbf{x}_N] \in \mathbb{R}^{26 \times N}$, PCA learns a linear transform $\mathbf{D} \in \mathbb{R}^{K \times 26}$, $1 \leq K \leq 26$ to project them into a lower dimension space which has a maximum variance. K is called the reduced dimension, which plays an important role in the follow-up classification step. Then the dimensionality reduction step is implemented

by $\bar{\mathbf{X}} = \mathbf{D}\mathbf{X} \in \mathbb{R}^{K \times N}$, where $\bar{\mathbf{X}}$ is the sample matrix after dimensionality reduction. Since the reduced dimensionality K is less than the number of samples, PCA can alleviate the problem of overfitting to a certain extent. After dimensionality reduction by PCA, the noise and redundant information in the samples will be reduced, which will lead to an improvement of subsequent classification accuracy. At the same time, reducing the dimensions of samples will inevitably lead to information loss. The more dimensions are reduced, the more information will be lost. Therefore, it is crucial to select an appropriate dimensionality K for subsequent experiments.

(3) Dataset balancing

Because dysphagia is a very widespread disorder, the collected samples are unbalanced, with a lot more patients than normal. It has been reported that when the machine learning model is trained with unbalanced data, the model will show a certain bias, preferring the majority classes. This is because the minority class instances rarely occur in the training process, and the model tends to predict the samples to the majority class to reduce the prediction error. As a result, test samples that fall into a minority class are more likely to be misclassified. For the detection of dysphagia, which is a binary classification task whose majority class is patients class, the model will show higher sensitivity and lower specificity.

The commonly used method is under-sampling technique. In the under-sampling, in order to balance the size of each class in the training process, we remove some samples of the majority class. Therefore, when a model is trained on balanced data, it should show a unbiased behavior. Random under-sampling is the simplest method, and its performance is similar to other complex methods. This paper uses the random under-sampling method to balance the training process and alleviate the uncertainty of the model.

2.3. Integrated machine learning classifiers

The detection of dysphagia can be viewed as a binary classification task. An appropriate classifier is very important to obtain good diagnostic accuracies. Generally, a single classifier is often not stable and robust enough. In order to combine the advantages of different classifiers, we integrate three widely used classifiers to make the final decisions. Those three classifiers are Support Vector Machine, Multi-Layer Perceptron, and Adaptive boosting.

(1) Support Vector Machine

Support vector machine (SVM) is one of the most robust prediction methods based on statistical learning frameworks, making it a non-probabilistic binary linear classifier. SVM maps the training samples to points in the space to maximize the gap between the two categories. Then new samples are then mapped into the same space and predicted to a category based on which side of the gap they fall. When dealing with the linear non-separable problems, SVM can effectively perform so-called kernel tricks by mapping its inputs to a higher-dimensional feature space. Billah and Waheed (2018) extracted features from endoscopic images, which are then used for training an SVM, and achieved an accuracy of 98.23%.

SVM performs classification tasks by constructing a hyper-plane in a higher dimensional space. Normally a good separation can be achieved by the hyper-plane, which has the largest distance to the nearest training data points of any class (so-called margin). Mathematically, assume that there are N training sample $\mathbf{x}_n \in \mathbb{R}^M$, $1 \leq n \leq N$, with their corresponding class labels $y_n \in \{+1, -1\}$. The standard SVM finds a projection vector $\mathbf{w} \in \mathbb{R}^M$ and a bias $b \in \mathbb{R}$ by a constrained optimization problem as follows:

$$\begin{aligned} \min_{\mathbf{w}, b, \xi} \quad & \frac{1}{2} \mathbf{w}^T \mathbf{w} + C \sum_{n=1}^N \xi_n \\ \text{s. t.} \quad & y_n (\mathbf{w}^T \mathbf{x}_n + b) \geq 1 - \xi_n, 1 \leq n \leq N, \xi_n \geq 0, \end{aligned} \quad (1)$$

Table 3

Details of extracted features from each sample by Praat software. M: Average of each feature values, S: standard deviation of each feature values. Health: group of healthy samples in the training set; Dysphagia: group of dysphagia samples in the training set.

Feature number	Features	Health	Dysphagia
	Description	M±S	M±S
1	Jitter(local)	0.30377±0.26590	0.35173±0.31327
2	Jitter(local,absolute)	18.93314±21.52490	21.93552±25.33179
3	Jitter(rap)	0.09380±0.11975	0.11178±0.13412
4	Jitter(ppq5)	0.15531±0.17312	0.18210±0.19262
5	Jitter(ddp)	0.28141±0.35916	0.33533±0.40232
6	Shimmer(local)	5.77858±3.32449	6.36761±3.90379
7	Shimmer(local,dB)	0.54328±0.31014	0.59791±0.35358
8	Shimmer(apq3)	3.04417±1.77977	3.29508±2.08805
9	Shimmer(apq5)	3.48237±2.06848	3.81671±2.35559
10	Shimmer(apq11)	4.1976±2.57634	4.87585±3.02641
11	Shimmer(dda)	9.13245±5.33939	9.88522±6.26406
12	Pitch(median)	198.62322±60.08816	199.75590±56.77805
13	Pitch(mean)	194.63906±55.44313	198.13517±53.39967
14	Pitch(standard deviation)	21.88796±20.18672	21.94387±20.33164
15	Pitch(minimum)	133.18807±59.52207	139.69190±58.05125
16	Pitch(maximum)	247.43522±75.67653	253.91390±81.10927
17	Period(number of pulses)	1188.86747±513.88004	235.18881±616.08294
18	Period(number of periods)	1186.46988±513.76281	232.35664±616.91059
19	Period(mean period)	0.00555±0.00157	0.00543±0.00151
20	Period(standard deviation)	0.00074±0.00077	0.00073±0.00079
21	Miscellaneous(Auto-correlation)	0.95992±0.04575	0.95361±0.05537
22	Miscellaneous(Noise-to-Harmonic)	0.05205±0.06532	0.06322±0.08955
23	Miscellaneous(Harmonic-to-Noise)	18.78390±5.11832	18.21039±5.54148
24	Miscellaneous(Fraction of locally unvoiced frames)	0.00424±0.00917	0.01292±0.04298
25	Miscellaneous(Number of voice breaks)	0.83133±1.93770	1.20979±3.63328
26	Miscellaneous(Degree of voice breaks)	0.00685±0.01741	0.01613±0.05518

where $\xi = [\xi_1; \xi_2; \dots; \xi_N]$ is a vector which contains all the slack variables, and ξ_i is also called the marginal error for the i th sample. This is called the primal problem. By solving for the Lagrangian dual of the above problem, the dual problem can be obtained as:

$$\begin{aligned} \max_{\alpha} \quad & \sum_{n=1}^N \alpha_n - \frac{1}{2} \sum_{n_1=1}^N \sum_{n_2=1}^N \alpha_{n_1} \alpha_{n_2} y_{n_1} y_{n_2} \mathbf{x}_{n_1}^T \mathbf{x}_{n_2} \\ \text{s. t.} \quad & \sum_{n=1}^N \alpha_n y_n = 0, \alpha_n \geq 0, 1 \leq n \leq N. \end{aligned} \quad (2)$$

It is a quadratic programming problem, which have some efficient solvers.

In order to overcome the limitation of linear SVM, we adopt the kernel method. It maps the original space to a higher dimensional feature space by $\phi(\cdot)$ and makes the sample linear separable in this sample space. Moreover, the kernel function κ should satisfy: $\kappa(\mathbf{x}_i, \mathbf{x}_j) = \phi(\mathbf{x}_i)^T \phi(\mathbf{x}_j)$. Once the optimization problem is solved, the predicted label y_{pred} of decision function for a given sample $\mathbf{x}$ becomes:

$$y_{\text{pred}} = \text{sign} \left(\sum_{n=1}^N \alpha_n y_n \kappa(\mathbf{x}, \mathbf{x}_n) + b \right). \quad (3)$$

(2) Adaptive boosting

Boosting classifier is composed of multiple weak classifiers, which predicts an unknown sample with each weak classifier respectively, and then votes to get the result. During training, each weak classifier is trained in turn, and each weak classifier focuses on the samples misclassified by the previous weak classifier. Weak classifier is a very simple classifier with small computation and low precision. In the basic Adaboost algorithm, each weak classifier has a weight, and the weighted sum of the prediction results of the weak classifier forms the final prediction result. During training, training samples also have weights, which are dynamically adjusted in the training process. Samples misclassified by the previous weak classifier will increase the weight, so the algorithm will pay attention to samples that are difficult to be divided. Compared with other classifiers, it adjusts the classifier's performance based on error feedback during testing. The final classifier for Adaboost is formulated as:

$$H(\mathbf{x}) = \text{sign} \left(\sum_{k=1}^K \alpha_k h_k(\mathbf{x}) \right), \quad (4)$$

where α_k is the weight and $h_k(\mathbf{x})$ is the prediction result for the k -th weak classifier.

Suppose we have N training samples $\mathbf{x}_n, 1 \leq n \leq N$, where $\mathbf{x}_n \in \mathbb{R}^M$, with the corresponding labels $y_n \in \{+1, -1\}$. Before the Adaboost algorithm starts, the weight distribution of the training data $D(k) = (w_{k,1}, w_{k,2}, \dots, w_{k,N})$ is initialized. Each training sample is initially assigned the same weight as:

$$w_{1,n} = \frac{1}{N}, n = 1, 2, \dots, N. \quad (5)$$

Adaboost algorithm will carry out multiple iterations, and each iteration can be roughly divided into three steps. Firstly, all the base classifiers are trained based on the weight distribution of the previous one, and the classification error rate of these classifiers on the training set is calculated as:

$$e_k = P(h_k(\mathbf{x}_n) \neq y_n). \quad (6)$$

Secondly, the weight of each weak classifier is updated as follows:

$$\alpha_k = \frac{1}{2} \log \frac{1 - e_k}{e_k}. \quad (7)$$

The weight represents the importance of the final corresponding classifier in the final classifier. It can be seen from the above equation that the weight increases with the decrease of the error, which means that the basic classifier with a smaller classification error rate plays a greater role in the final classifier.

$$w_{k+1,n} = \frac{w_{k,n}}{Z_k} \exp(-\alpha_k y_n h_k(\mathbf{x}_n)), \quad (8)$$

where Z_k is a normalization factor. Finally, the weight distribution of the training data is updated and used in the next iteration. At the end of all iterations, we combine the final classifier as shown in Formula (4).

(3) Multi-Layer Perceptron

A multi-layer perceptron (MLP) is a class of feedforward artificial neural network. A perceptron uses its input weights to form a linear combination, which in turn produces a single output through a non-linear activation function. The output (denote as a neuron) can be formulated as:

$$y = \phi \left(\sum_{n=1}^N \mathbf{w}_n \mathbf{x}_n + b \right) = \phi(\mathbf{w}^T \mathbf{x} + b), \quad (9)$$

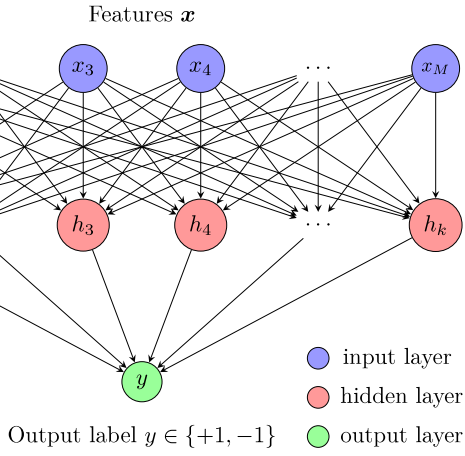


Fig. 4. One hidden layer MLP.

where $\mathbf{w}$ denotes the weights, $\mathbf{x}$ is the input vector and $\phi(\cdot) : \mathbb{R} \rightarrow \mathbb{R}$ is a non-linear activation function, set by default as:

$$\phi(x) = \frac{1 - \exp(-2x)}{1 + \exp(2x)}. \quad (10)$$

An MLP consists of multiple perceptrons. It consists of an input layer to receive the input vector, an output layer that makes a prediction about the input vector, and an arbitrary number of hidden layers in between those two layers. All the layers are full-connected to each other. These hidden layers are the real computational engine of MLP, and they allow MLP to approximate any continuous functions. Fig. 4 gives an illustration of an MLP with only one hidden layer.

(4) Integrated Classifier

We propose a new idea on combining conceptually different machine learning classifiers to predict the class labels based on the MLP classifier. MLP classifier can emphasize the features that have better discrimination ability than others in a feature set by increasing their weights. This theory is helpful in combining the advantages of different classifiers. First, the base classifiers are trained independently and predict the labels of the training samples. Second, the prediction results calculated from each base classifier are adopted as one of the input features of the MLP classifier to train the integrated model. If the classification performances of the base classifiers are poor, MLP will ignore their prediction results or give them small weights. If the classification performances of the base classifiers are good, it will help the new integrated MLP classifier make a better decision. The code of the programs and the calculated feature sets are available at <https://github.com/hxhj555/Integrated-Classifier-for-Dysphagia-Detection>.

Based on this theory, such integration is useful for emphasizing the well-performing base classifiers to improve the overall performance of the integrated model. However, the integration of different classifiers increases the computing complexity of the classification task. We must calculate the results of each base classifier and design some rules to combine their performances, which causes additional time cost of the classification task.

3. Experimental results

In this section, the experimental results of the proposed classification approach are calculated and discussed. First, we introduce three evaluation metrics that are adopted to verify the effectiveness of the proposed classifier. Second, the details of the established classification model are described. Finally, we give the classification results and analysis.

Table 4

Participant Characteristics of the Health, and Dysphagia Groups. SD denotes standard deviation (Wang et al., 2021).

Characteristic	All	Health	Dysphagia
Female+Male	226	83	143
Female	144	48	96
Male	82	35	47
Age (years, mean $\pm$ SD)	84.2 $\pm$ 5.8	83.3 $\pm$ 5.3	84.7 $\pm$ 5.6

3.1. Evaluation metrics

In this experiment, we use three metrics named as accuracy Acc, sensitivity Sen, and specificity Spe, to describe the performance of the classifier. The three metrics are defined as follows:

$$\text{Acc} = \frac{N_{TP} + N_{TN}}{N_{TP} + N_{FP} + N_{TN} + N_{FN}}, \quad (11)$$

$$\text{Sen} = \frac{N_{TP}}{N_{TP} + N_{FN}}, \quad (12)$$

$$\text{Spe} = \frac{N_{TN}}{N_{FP} + N_{TN}}. \quad (13)$$

In Eqs. (11), (12) and (13), N_{TP} and N_{TN} represent the number of patients with dysphagia and normal subjects which are correctly classified. N_{FP} and N_{FN} represent the number of misclassified patients and normal persons respectively. Accuracy describes the proportion of the sample that is correctly classified to the total sample. Sensitivity is the percentage of people with dysphagia that is correctly detected. Specificity is the proportion of normal persons who are diagnosed as normal.

Thanks to National Clinical Research Center for Geriatrics (West China Hospital of Sichuan University) for providing us with the experimental dataset. The dataset included throat signals from 226 volunteers. Table 4 shows the details of the dataset. The data collection process has passed the medical ethics review, which was approved by Bio-medicine Ethics Committee of West China Hospital, Sichuan University with the Registration number ChiCTR1900023625. All experimental data are collected using a laryngeal vibrator and stored as audio data. All the samples are labeled by medical doctor.

3.2. Models' setting

In this experiment, 75% of the data are used for training the classifier, and 25% of the data are used for testing the classifier. The base classifiers are trained independently using the same training set, and they work in parallel before the integration. The prediction results and the original speech feature sets of the training samples are all adopted to train the integrated MLP classifier. For each test sample, its prediction results by the base classifiers and its speech feature set are input to the integrated classifier to obtain the output.

The hyper parameters of these three classifiers are set by considering the complexity of this classification task. For the AdaBoost classifier, the decision tree is selected as the base estimator. The number of estimators and the learning rate is set to 300 and 0.2, respectively. The maximum depth of the tree is 2 and the minimum number of samples in each leaf node is 5. For the SVM classifier, we select the radial basis function kernel and set the hyper parameter C as 14.3142. This hyper parameter is determined by testing the C from 0.1 to 20 (50 evenly divided values) using 5-fold cross validation. For the MLP classifier, the hidden layer size is (35, 19) and the optimizer is the SGD. The hyperbolic tan function is adopted as the activation function. These hyper parameters influence the convergence conditions for each base classifier. In our work, the base classifiers are integrated by the MLP classifier. The iteration times of the MLP classifier are the main factor that decides on the convergence of the model.

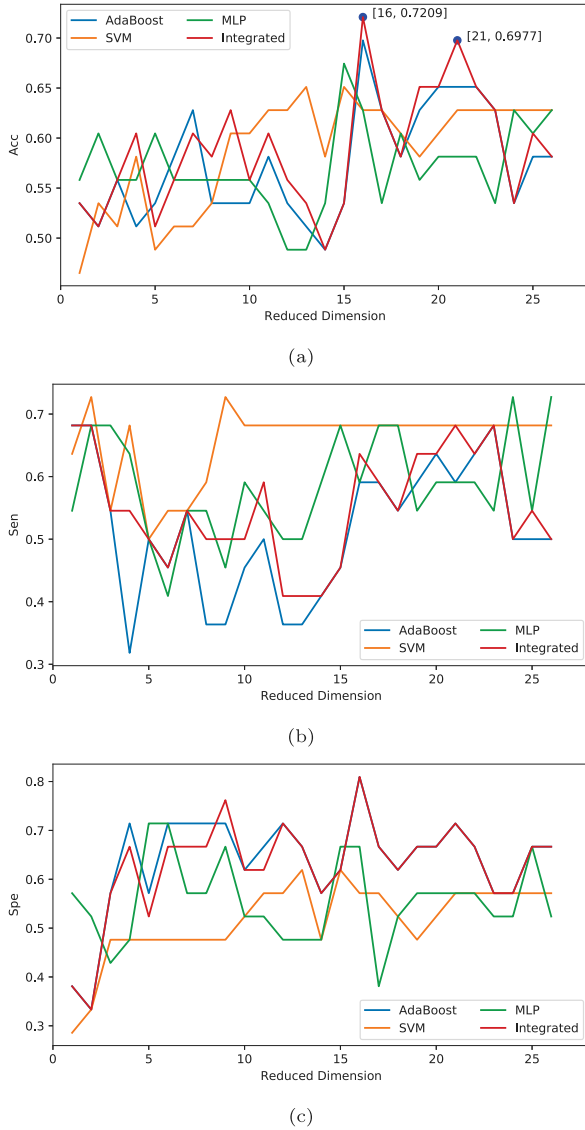


Fig. 5. The performance of three single classifiers and the integrated classifier when the reduced dimension K takes the value from 1 to 26. (a) The accuracy metric; (b) The sensitivity metric; (c) The specificity metric.

3.3. Classification results

In this subsection, 26 groups of classification experiments are conducted based on different dimensions of feature set after PCA. The purpose of doing so many groups of experiments is twofold. First, from the perspective that PCA transform will have an impact on the performance of the classifier, we need to do multiple groups of experiments to select the most appropriate dimension K to get better performance. Second, from the point of view that a new sample set will be obtained after PCA transformation, we can conduct multiple groups of experiments to verify the effectiveness of our ensemble classifier on different sample sets. Based on the above two aspects, we conducted 26 groups of experiments to traverse K from 1 to 26. In these experiments, the settings of each base classifier remain the same. The results of the three related metrics, Acc, Sen and Spe that are changed with K are shown in Fig. 5.

In Fig. 5(a), it shows that the integrated MLP classifier has better classification accuracies than each single classifier in most situations. With the increasing of the dimensions K , the integrated MLP classifier

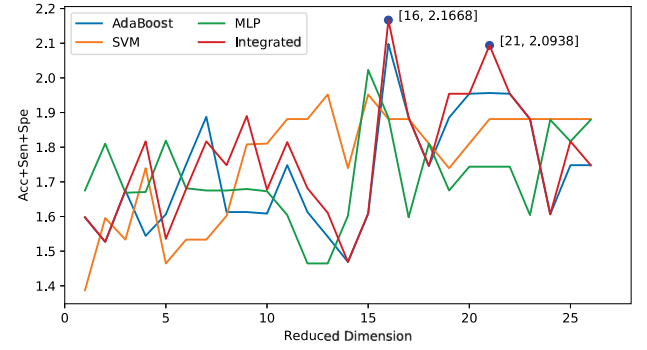


Fig. 6. The overall performance of three single classifiers and the integrated classifier with different K in terms of accuracy, sensitivity, and specificity.

Table 5

Performance of the integrated classifier among the 26 groups of experiments.

Evaluation	Bad	Acceptable	Good
Rate	0%(0)	57.7%(15)	42.3%(11)

obtains the best results when the reduced dimension K is 16. The classification accuracy reaches up to 72.09%. The specificity and sensitivity are 80.95% and 63.64%, respectively. Fig. 5(b) exhibits the results of sensitivity in these classification tasks. It is observed that SVM classifier has higher values of sensitivity than the other classifiers. However, the specificity of SVM classifier is poorer than all other classifiers as shown in Fig. 5(c).

These results in Fig. 5 also indicate that such integration can improve the performance of the original model. In this work, we integrate the classification model based on the MLP classifier. In order to verify the effectiveness of this integration method, the hyper parameters of the integrated MLP model remain the same with the original MLP classifier. As the results in these figures, the green lines draw the results of the original MLP classifier. The red lines show the results of the integrated MLP classifier. It exhibits that the integrated MLP classifier performs better than the original classifier.

In clinical practice, sensitivity and specificity need to be balanced as much as possible. However, we can know from Eqs. (11), (12) and (13), that given a certain accuracy, a larger value of one metric means a smaller value of the other metric. To take an extreme example, if the classifier predicts all the testing samples as dysphagia, the sensitivity would be 100%, but meanwhile the specificity would be 0%. The classification results of SVM classifier in Fig. 5(b) and (c) are a typical example. Such situations should be avoided. So the difference between the two metrics is also worth paying attention to. We can see that the sensitivity and specificity of the integrated classifier are relatively balanced in Fig. 5. This is because the integrated classifier is robust to a certain extent and does not lead to too much difference between the two metrics.

In order to compare the comprehensive results, we add up the Acc, Sen, and Spe with different K in each figure. Fig. 6 shows the final comparison. We conclude that the selection of the reduced dimension K has an important impact on the performance of the classifier. By selecting an appropriate K , the final classification accuracy can be greatly improved.

It shows that in 26 groups of experimental species, the result of integrated classifier is not the best in all cases. In order to evaluate the stability of the integrated MLP classifier, we divide all the results into three categories. For a same K , denote the accuracy of three single classifiers and the integrated classifier as $\text{Acc}(h_i), i = 1, 2, 3, \text{Acc}(H)$, respectively. Among all 26 K values, mark the result of (14) as 'Good', mark the result of (15) as 'Bad' and others are marked as 'Acceptable'.

$$\text{Acc}(H) \geq \max \{ \text{Acc}(h_i), i = 1, 2, 3 \} \quad (14)$$

Table 6
Classification results of the methods in existing work.

References	Data sets	Feature sets	Acc	Sen	Spe
Santamato et al. (2009)	15 dysphagia patients; 60 healthy controls	DSS, PF (peak frequency), PI (peak intensity)	53.49%	54.55%	52.38%
López-De-Ipiña et al. (2016)	5 patients with PD; 10 healthy controls	HNR, pitch, PE (permutation entropy)	55.81%	72.73%	38.10%
López-De-Ipiña et al. (2018)	6 patients with PD; no healthy controls	Classical features (CF): HNR, jitter, shimmer, pitch, etc.	53.49%	54.55%	52.38%
		Non-linear features (NLF): PE, Shannon, fractal dimension	51.16%	59.09%	42.85%
Roldan-Vasco et al. (2021)	46 dysphagia patients; 46 healthy controls	Phonation features on vowels (jitter, shimmer, F0, mfcc, etc.)	58.14%	59.09%	57.14%
Ours	143 dysphagia patients; 83 healthy controls	Jitter, shimmer, pitch, Period, Miscellaneous	65.12%	68.18%	57.14%

$$\text{Acc}(H) < \min \{ \text{Acc}(h_i), i = 1, 2, 3 \} \quad (15)$$

Table 5 shows that the integrated classifier can guarantee an acceptable result with a high probability (57.7%), and in some cases (42.3%) it can get better results than the single classifier. Thus we conclude that, for the classification tasks where it is difficult to determine which classifier to use, the integration method provides a solution which is stable in many cases.

In recent studies, some researchers have successfully applied and proposed some new ideas for designing the decision-making classifiers. Chiang, Shih, Lin, and Shih (2014) adopt an APN model to transform the decision systems into simple sprouting trees and achieve the automatic classification of Arrhythmic beat signals. APN is a powerful method for representing knowledge and logic reasoning in the domain of decision support systems. Albu, Precup, and Teban (2019) introduce several applications of Artificial Neural Networks method, such as skin diseases diagnosis, Hepatitis B prediction, and Stroke risk prediction. They report that some types of AI tools are adopted in clinical situations, and the computer-aided decision-making systems still have enough room for improvements. Upadhyay and Nagpal (2020) adopt the SVM and Radial Basis Function Neural Network (RBFNN) to classify stress conditions of sleep EEG. Their results demonstrate that RBFNN and SVM models may be successfully employed as decision support tools in clinical studies. These researches indicate that there are many strategies for improving and designing new classifiers. It is meaningful to contribute more efforts to exploring effective machine learning methods.

3.4. Comparative experiments

In this subsection, we conduct the comparative experiments with the existing methods. Some of the previous works have achieved automatic detection of dysphagia based on the analysis of speech signal or swallowing sounds. The feature sets in these studies are tested using the same SVM classifier and PCA technique adopted in this work. The classification results of these features are listed in Table 6.

As the results listed in Table 6, all the feature sets are tested by the same classifier and data set in our work. It is shown that the group of features in our work obtains the best results. The second occurs in research (Roldan-Vasco et al., 2021). The group of features in our work could be a reference to the future studies. In addition, our experiments are carried out based on a larger number of samples than those in the similar research. A larger number of samples can help explore more effective and generalized speech features and classification model.

4. Conclusion

In this paper, we propose and implement an integrated classifier that can be used to detect dysphagia based on throat signals collected from subjects. This integrated classifier combines the classification advantages of three classifiers AdaBoost, SVM, and MLP to make the

final prediction. We perform some experiments to demonstrate the effectiveness of the proposed method.

For the first time, we use a refitted throat vibrator instead of a traditional microphone to capture the speech information in audio data acquisition. In feature engineering, we perform the feature extraction on the audio signals by Praat software. To improve the performance of the classifier, data normalization is performed on the feature vector; PCA is adopted to perform dimensionality reduction step; then random under-sampling is adopted to deal with the unbalanced training data. After a brief introduction for three classifiers, we give the integration method to combine three classifiers. Finally, we evaluate the trained model with test subset. Experimental results show that the integrated classifier provide a better performance than the single classifier and is robust in many cases.

Though this model performs well in this work, some factors might degrade its performance. Since this model is established based on the speech data collected from older adults, it might perform poorly in young dysphagia patients because some speech features would change with age. In addition, the selection of the base classifiers' hyper parameters also influences its performance, such as the penalty factor of the SVM classifier, the number and learning rate of the ensemble classifier, and the size of the hidden layer in MLP classifier. These hyper parameters should be optimized in the model training process. In the future, we intend to introduce other modal data together to develop a better system to support medical diagnosis.

CRedit authorship contribution statement

Hengling Zhao: Conceptualization, Writing – original draft, Resources. **Yangyang Jiang:** Investigation, Data curation, Resources. **Shenghan Wang:** Methodology, Software. **Fei He:** Methodology, Writing – review & editing. **Fangzhou Ren:** Investigation, Resources. **Zhonghao Zhang:** Investigation, Resources. **Xue Yang:** Data curation. **Ce Zhu:** Investigation. **Jirong Yue:** Resources. **Ying Li:** Supervision, Conceptualization. **Yipeng Liu:** Supervision, Conceptualization, Writing – review & editing.

Declaration of competing interest

No author associated with this paper has disclosed any potential or pertinent conflicts which may be perceived to have impending conflict with this work.

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